

organic spintronics, creating devices that utilise the spin of electrons to read, write and manipulate data.

Ezekiel Johnston-Halperin, assistant professor of physics, and his team, combined gallium arsenide (an inorganic semiconductor) with a unique plastic material that is under development in colleague Arthur J. Epstein's lab at Ohio State University. They transmitted a spin-polarised electrical current from the plastic material, through the gallium arsenide, and into a light-emitting diode (LED) as proof that the organic and inorganic parts were working together.

"Hybrid structures promise functionality that no other materials, neither organic nor inorganic, can currently achieve alone. We have opened the door to linking this exciting new material to traditional electronic devices with transistor and logic functionality. In the longer term, this work promises new, chemical based functionality for spintronic devices," said Johnston-Halperin.

Scan your face in 3D with your iPhone

A new iPhone app called Trimensional turns your face or other object into a 3D computer model that can easily be edited or even reproduced with a 3D printer.

Created by Grant Schindler, a computer scientist at the Georgia In-

stitute of Technology, the app uses the iPhone's screen to project four different patterns of light onto the object being scanned and snaps photos with the front-mounted camera. It then compares the lighting patterns and stitches the photos together to produce the 3D model.

Users can also e-mail animated videos of their 3D model or export their creation to CAD programs or 3D applications such as Maya.

'3D towers' to double disk storage capacity

A team of researchers at SPINTEC, in Grenoble, France, has figured out a way to double the areal density of information by essentially cutting the magnetic media into small pieces and building a 3D tower out of it. This enhances the amount of data that can be stored in a magnetic storage device and provides a method to reach beyond a wall of physical limits that the currently used technology is hitting.

"Our new approach involves using bit-patterned media, which are made of arrays of physically separated magnetic nanodots, with each nanodot carrying one bit of information. To further extend the storage density, it's possible to increase the number of bits per dots by stacking several magnetic layers to obtain a multilevel magnetic recording device," explained Jerome Moritz, a

researcher at SPINTEC.

The researchers were able to demonstrate that the best way to achieve a two-bit-per-dot media involves stacking in-plane and perpendicular-to-plane magnetic media atop each dot. The perpendicularly magnetised layer can be read right above the dot, whereas the in-plane magnetised layer can be read between dots. This enables doubling of the areal density for a given dot size by taking better advantage of the whole patterned media area.

Hide data in hard drive without encryption

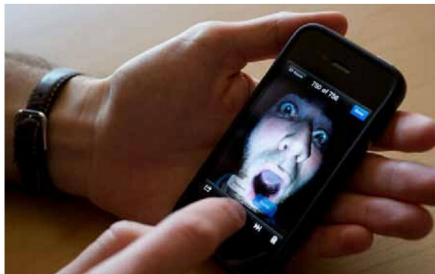
A new software developed by Hassan Khan and colleagues at the National University of Science and Technology in Islamabad, Pakistan, lets you hide data on a hard drive without using encryption.

Instead of using a cipher to scramble text, the method uses 'steganography'—hiding data in plain sight—to manipulate the location of data fragments. It takes advantage of the manner in which hard drives store file data in numerous small chunks called 'clusters.' The software ensures that the clusters of a file are positioned according to a code. All that a person at the other end needs to know is which file's cluster positions have been encoded.

The code depends on the location of the sequential clusters in the file on the hard drive. If they are adjacent, this corresponds to a binary 1 in the secret message. If they are stored at different places on the disk, this encodes a binary 0. The recipient then uses the same software to get the file's cluster positions, and hence the message.

Browser to decide what you share with websites

Many websites ask visitors to share their personal information in exchange for receiving useful personalised services. Microsoft has come up with a Web browser and associated protocols that could strengthen the user's hand



Trimensional 3D scanner iPhone app